Tutorial 9(Unit -3) Integration **Faculty Signature: ________________**

Name: ___________________________ **Roll No: ____________________**

Date: ____________________________ **Marks: ______**

Evaluate the following:

1. $\int (2x^2 + e^x) dx$ Ans. $\frac{2}{3}x^3 + e^x + C$

2. $\int y^2 \sqrt[3]{y}\, dy$ Ans. $\frac{3y^{\frac{10}{3}}}{10} + C$

3. $\int \frac{x^3 - x^2 + x - 1}{x - 1}\, dx$ Ans. $\frac{x^3}{3} + x + C$

4. $\int (2x - 3\cos x + e^x)\, dx$ Ans. $x^2 - 3\sin x + e^x + C$

5. $\int \frac{x^3 + 3x + 4}{\sqrt{x}}\, dx$ Ans. $\frac{2}{7}x^{7/2} + 2x^{3/2} + 8\sqrt{x} + C$

6. $\int \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2\, dx$ Ans. $\frac{x^2}{2} + \log(x) - 2x + C$

7. $\int (1 - x)\sqrt{x}\, dx$ Ans. $\frac{2}{3}x^{\frac{3}{2}} - \frac{2}{5}x^{\frac{5}{2}} + C$

8. $\int \left(2x^2 - 3\sin x + 5\sqrt{x}\right) dx$ Ans. $\frac{2}{3}x^3 + 3\cos x + \frac{10}{3}x^{\frac{3}{2}} + C$

Tutorial 10(Unit -3) Integration **Faculty Signature: ________________**

Name: __________________________ **Roll No: ___________________**

Date: ___________________________ **Marks: ______**

Evaluate the following:

1. $\int(\sin 2x - 4e^{3x})\, dx$ Ans. $\frac{-1}{2}\cos 2x - \frac{4}{3}e^{3x} + C$

2. $\int \frac{2-3\sin x}{cos^2 x}\, dx$ Ans. $2\tan x - 3\sec x + C$

3. $\int \sin 4x . \sin 8x\ \ dx$ Ans. $\frac{1}{2}\left[\frac{1}{4}\sin 4x - \frac{1}{12}\sin 12x\right] + C$

4. $\int\left(\frac{2-3\sin x}{\cos^2 x}\right)dx$ Ans: $2\tan x - 3\sec x + c$

5. $\int\frac{1}{16+y^2}dy$ Ans: $\frac{1}{4}\tan^{-1}\frac{y}{4}+c$

6. $\int\frac{1}{3x+4}dx$ Ans: $\frac{1}{3}\log(3x+4)+c$

7. If $\frac{d}{dx}\left(f(x)\right) = 4x^3 - \frac{3}{x^4}$ such that $f(2)=0$, find $f(x)$

Ans: $x^4 + \frac{1}{x^3} - \frac{129}{8}$

Tutorial 11(Unit -3) Integration **Faculty Signature: ________________**

Name: ___________________________ **Roll No: ___________________**

Date: ____________________________ **Marks: ______**

Evaluate the following:

1. $\int \cos 2x \, . \cos 4x \, . \cos 6x \;\, dx$ Ans. $\frac{1}{4}\left[\frac{1}{12}\sin 12x + x + \frac{1}{8}\sin 8x + \frac{1}{4}\sin 4x\right] + C$
2. $\int \sin x \, . \sin 2x \, . \sin 3x \;\, dx$ Ans. $\frac{1}{4}\left[\frac{1}{6}\cos 6x - \frac{1}{4}\cos 4x - \frac{1}{2}\cos 2x\right] + C$
3. $\int x . \sin 3x \;\, dx$ Ans. $\frac{-x}{3}\cos 3x + \frac{1}{9}\sin 3x + C$
4. $\int x . \tan^{-1} x \;\, dx$ Ans. $\frac{x^2}{2}\tan^{-1} x - \frac{x}{2} + \frac{1}{2}\tan^{-1} x + C$
5. $\int e^x (\sin x + \cos x) \, dx$ Ans. $e^x \sin x + C$
6. $\int (x^2 + 1) \log x \;\, dx$ Ans. $\left(\frac{x^3}{3} + x\right)\log x - \frac{x^3}{9} - x + C$
7. $\int e^x \left(\frac{1}{x} - \frac{1}{x^2}\right) \, dx$ Ans. $\frac{e^x}{x} + C$